



Monte Carlo for GNC Validation on Liquid Propelled Rockets

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Chapter 1

General Outline

The purpose of this document is to outline the Monte Carlo methods used for Verification and Validation (V&V) of Guidance, Navigation, & Control (GNC) algorithms for the TOAD Lander vehicle, with a specific focus on the vibration sensibility of these sensors. We will also cover the different sources and assumptions used to obtain distributions for these parameters.

This document will soon be updated to contain characterization procedures with a vibration table for our current MEMS IMU setup after promising results during ASTRAv2 testing.

Chapter 2

Vibration Rectification Bias in MEMS Sensors from LRE Vibration Regimes

2.1 Common Sources of bias in MEMS

Microelectronic Mechanical Sensors (or MEMS for short), are the most common type of customer-grade sensors, used every day in cellphones, consumer drones, and generally low stakes environments. These sensors function, in essence, as a tuned mass spring damper, and make use of the Coriolis effect (for gyroscopes) to obtain measurements of the parameters outlined.

A commonly talked about effect with MEMS gyroscopes especially is their bias, and especially their bias drift. Since these sensors are purely mechanical, the "springs" they use have a strong temperature dependence, and due to thermal contraction or expansion, an inherent drift can appear. Other sources of drift include "Rate Random Walk" (RRW).

TOAD's Navigation system is a GPS-aided Inertial Navigation System, which makes use of a Multiplicative Extended Kalman Filter to fuse accelerometer, gyroscope, magnetometer, & GPS data when the vehicle is on the pad to produce initial estimates of all these biases and initial conditions. However, due to the direct input feed-through into the accelerometer measurements in-flight, as well as the weak confidence in magnetometer data in flight from high current draws from valves or mis-calibrated sensors, gyroscope data accuracy is integral to our success. Now, the high-precision velocity data from our RTK system provides enough information for the filter to infer yaw and pitch values with high observability. The roll, however, remains unobservable to the GPS under vertical-flight conditions, and poorly observable under motion.¹ Our flight stability then depends on the roll error not exceeding a stability threshold.

Now, with proper calibration & thermal soak procedures for the board, our current IMU selection (the consumer grade ICM 40609-D) would give enough stability to provide a full 60 second flight under stable conditions via dead-reckoning, but an often overlooked source of drift for low-end MEMS IMU's is an effect called Vibration Rectification Error (VRE). This parameter is especially

¹A proof of this dynamic observability is available in the ASTRAv2 White-paper

relevant under the vibration loads from a liquid rocket engine, as is our case.

2.2 What is Vibration Rectification [VRE]?

Vibration Rectification is the primary troublemaker for MEMS sensors under high vibration loads. It is a direct consequence of the nonlinearity of mechanical sensors [1]:

Let us model an accelerometer ² as an n-th order polynomial. The output acceleration can be written as a function of the input (actual) acceleration:

$$a_{out} = K_0 + K_1 a_{in} + K_2 a_{in}^2 + K_3 a_{in}^3 \dots K_n a_{in}^n \quad (2.1)$$

It can be seen than when an input acceleration of the form

$$G_0 \cos(\omega t)$$

is inputted, the model morphs into:

$$a_{out} = K_0 + K_1 [G_0 \cos(\omega t)] + K_2 [G_0 \cos(\omega t)]^2 + \quad (2.2)$$

$$K_3 [G_0 \cos(\omega t)]^3 \dots [G_0 \cos(\omega t)]^n \quad (2.3)$$

Now let us look at the time averaged output of the function. We note that over time, all odd powered terms average out to zero, while the even terms do not:

$$\frac{1}{T} \int_0^T \cos^2(\omega t) = \frac{1}{2} \qquad \frac{1}{T} \int_0^T \cos^4(\omega t) = \frac{3}{8} \quad (2.4)$$

Substituting these terms back into our equation:

$$\bar{a}_{out} = K_0 + 0 + \frac{1}{2} K_2 G_0^2 + 0 \dots \quad (2.5)$$

To first order then, we're left with:

$$VRE = 0 - \bar{a}_{out} - K_0 \approx K_2 G_{RMS}^2 \quad (2.6)$$

K_2 is thus defined as our G^2 sensitivity. In essence, this derivation tells us that an IMU will develop an induced bias approximately proportional to the square of the RMS of the vibrations. Later on we will develop the expected G_{RMS} during flight, but one can expect anywhere from 4 to 5 G's of vibrations during flight, so without proper damping & sensors it is clear how quickly the accuracy can be degraded during flight.

²This document mostly focuses on VRE for gyroscopes, but this derivation is shown for an accelerometer. The effect is identical nevertheless, just different scaling.

Chapter 3

Monte Carlo Simulation & Parameter Disturbances

3.1 Sources & Assumptions for Baseline values

In an effort to begin validation and verification of our different GNC algorithms previous to hardware availability, we developed a Monte Carlo framework using our custom 6DoF simulation for TOAD. This simulation includes actuator models, sensor models including VRE, thermal drift, 2nd order dynamics, and position relative to vehicle CG, as well as modeling wind as a Gauss Markov Process. We have chosen to inject uncertainty into multiple parameters, a summary of which is available below:

Parameter	Symbol	Distribution Type	Dispersion
Moment of Inertia	δJ_{ii}	Uniform	$\pm 10\%$ of J_{nom}
Lever Arm Disturbance	$\delta \mathbf{r}_{TB}$	Normal	$\sigma = 0.01$ m per axis
Gyro Noise Power	σ_{gyro}^2	Log-Normal	$\sigma \approx 5.87 \times 10^{-7}$
Vibration Amplitude	G_{RMS}	Uniform	3 to 8 G_{RMS}
Grommet Stiffness	K_{grom}	Normal	$\sigma = 15\%$ of K_{nom}
Grommet Damping	B_{grom}	Normal	$\sigma = 15\%$ of B_{nom}
G^2 Sensitivity	K_{g^2}	Uniform	0.002 to 0.1 / sec/G_{RMS}^2
Wind Gain	K_{wind}	Rayleigh	$\sigma \approx 0.162$
Wind Covariance	σ_{wind}^2	Uniform	0 to 3 m/s

Table 3.1: Monte Carlo simulation parameters, distribution types, and dispersions.

3.1.1 Driving G_{RMS} and Spectral Distribution of Vibrations

Since we have yet to obtain engine data or the subsequent modal analysis for the vehicle, we look to prior art to find a distribution of plausible values for our driving vibrations at the avionics section (prior to any damping solution for the Flight Computer).

Source	Range (Hz)	Total (G_{RMS})
<i>SpaceX Rideshare Payload User's Guide</i> [2]	20 – 2000	5.57
<i>Firefly Alpha Payload User's Guide</i> [3]	20 – 2000	4.9
<i>Astrobotic Xodiac Payload User's Guide</i> [4]	1 – 150*	N/A*

* *Xodiac does not define a standard random vibration G_{RMS} . Instead, for the 1 – 150 Hz frequency band, it specifies a maximum 3-sigma in-flight acceleration of $\pm 2.73G$ in the vertical Z-axis, and an incomplete range of spectral data. Nevertheless it is the closest vehicle in size to ours, so it is worth to consider.*

Table 3.2: Broadband ranges and total G_{RMS} values for various payload guides.

3.1.2 Damping Grommets

The Monte Carlo setup allows us to select multiple grommets to help damp out the vibrations from the vehicle. Given a specific Power Spectral Distribution of vibrations $S_{in}(f)$ (as outlined in the previous section), we can calculate how much is transmitted to the IMU. For a specific grommet, the natural frequency is defined by:

$$f_n = \frac{1}{2\pi} \sqrt{\frac{k}{m_{IMU}}} \quad (3.1)$$

where m_{IMU} represents the mass of the flight computer. We can now define a damping ratio given by $\zeta = \frac{c}{2\sqrt{km}}$. The 1DoF absolute transmissibility formula is defined by:

$$T(f) = \sqrt{\frac{1 + (2\zeta \frac{f}{f_n})^2}{(1 - [\frac{f}{f_n}]^2)^2 + (2\zeta \frac{f}{f_n})^2}} \quad (3.2)$$

The output PSD after damping is then simply:

$$S_{out}(f) = [T(f)]^2 \cdot S_{in}(f) \quad (3.3)$$

We calculate the output $G_{RMS,IMU}$ as:

$$G_{RMS,IMU} = \sqrt{\int_0^\infty S_{out}(f) df} \quad (3.4)$$

3.1.3 IMU G^2 Parameters

Since most (if not all) customer grade IMU's lack published parameters for G^2 sensitivity, we are forced to pick a high upper bound while we await Flight Computer vibration testing. To baseline our lower-bound, we look at the higher-end MEMS sensors available on the market for a reference:

Source	Gyroscope VRE	Approx. Price (USD)
<i>Analog Devices ADIS16488A Datasheet</i> [5]	$0.00014^\circ/\text{sec}/g^2$	420
<i>Analog Devices ADIS16470 Datasheet</i> [6]	$0.001^\circ/\text{sec}/g^2$	402
<i>SBG Systems Pulse-40 Hardware Manual</i> [7]	$0.000056^\circ/\text{sec}/g^2$	1000+

Table 3.3: Published Vibration Rectification Error (VRE) and price points for various MEMS IMUs.

Consumer-grade MEMS IMUs typically omit VRE from their datasheets because their performance degrades significantly under vibration, so to make sure we have an absolute maximum upper bound, we set it to $0.1^\circ/\text{sec}/g^2$.

3.2 Success Criteria

For the simulations, we established multiple success criteria the run must meet to be considered a success. These include challenge requirements from the Collegiate Propulsive Lander Challenge (CPLC), as well as vehicle requirements for tip angle, maximum structural loads from landing, among others.

Metric	Success Threshold
Flight Time	< 60 s
Hover Accuracy	Within 1 m ³ of hover point for 10 s [*]
Touchdown Tilt Angle	$< 15^\circ$ from vertical
Impact Velocity	< 0.2 m/s
Lateral Velocity at Approach (1 m above pad)	< 0.3 ft/s
Landing Error	< 2.0 m

^{*} As per CPLC challenge requirements.

Table 3.4: Simulation success criteria and required thresholds.

Chapter 4

Results

The presented results are outputs from a 300-run batch of simulation with the distributions presented before. It is worth noting that for the sake of realism, the magnetometer model included significant uncalibrated offsets: The final transformation matrix is defined as $M = dM + I_{3 \times 3}$:

Table 4.1: Summary of Soft Iron Magnetic Disturbances

Parameter	Value	Percentage	Description
dM_{xx}	0.035	3.5%	Scaling (affects X and Y axes)
dM_{zz}	0.060	6.0%	Scaling (affects Z axis)
dM_{xz}	0.010	1.0%	Cross-axis coupling (X-Z and Y-Z axes)
dM_{xy}	0.005	0.5%	Cross-axis coupling (X-Y axes)

$$M = \begin{bmatrix} 1.035 & 0.005 & 0.010 \\ 0.005 & 1.035 & 0.010 \\ 0.010 & 0.010 & 1.060 \end{bmatrix} \quad (4.1)$$

In addition to the soft iron scaling and coupling distortions modeled by matrix M , the system includes a hard iron offset.

$$\mathbf{B}_{meas} = M\mathbf{B}_{true} + \mathbf{b}$$

The hard iron bias vector is randomly generated such that its overall magnitude represents a 5% offset relative to the nominal magnetic field strength (e.g., $\|\mathbf{b}\| = 0.05 \times B_{nominal}$). This simulates permanent magnetization present in the vicinity.

It is also worth noting that since our trajectory & controller functions off of a Waypoint system, the Root Mean Squared Error (RMSE) metrics from the following plots are heavily inflated, and are meant to be used as a relative metric between simulation runs, not as an absolute measure of the error across the trajectory.

The "Stability Manifold" plots shown here are simply heatmaps of the lateral position RMSE & the filter error RMSE as a function of the vibration relevant parameters. The fact that there is no meaningful correlation between these indicates good compensation from the filter.

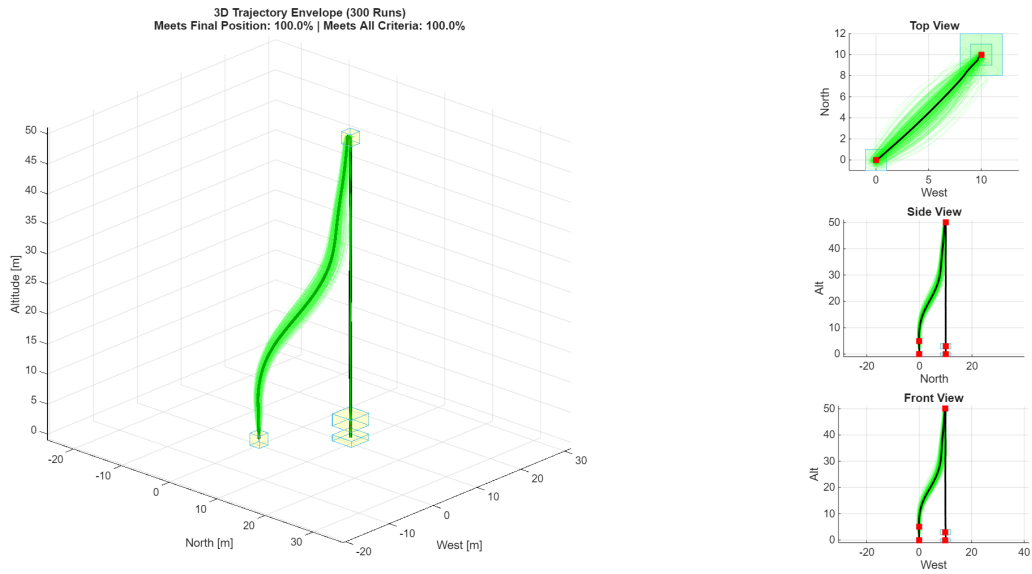


Figure 4.1: Mission Trajectories

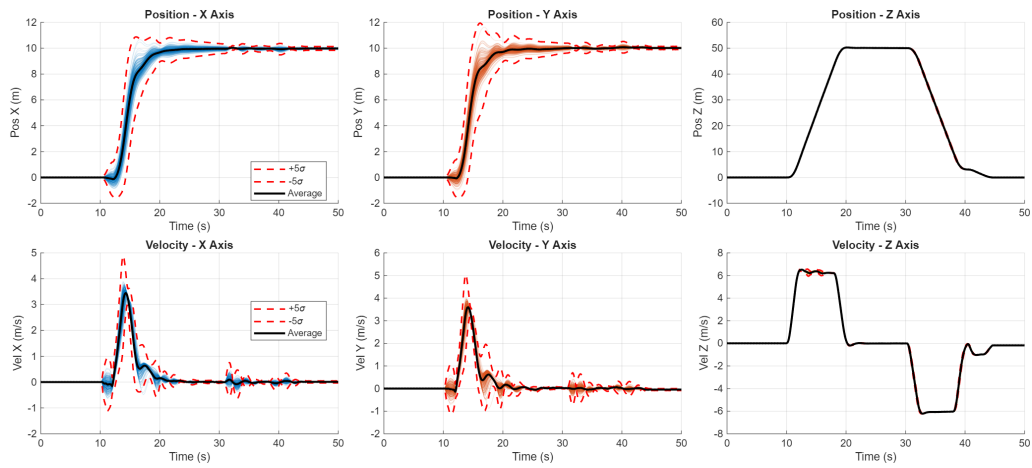


Figure 4.2: State Distributions (5-Sigma)

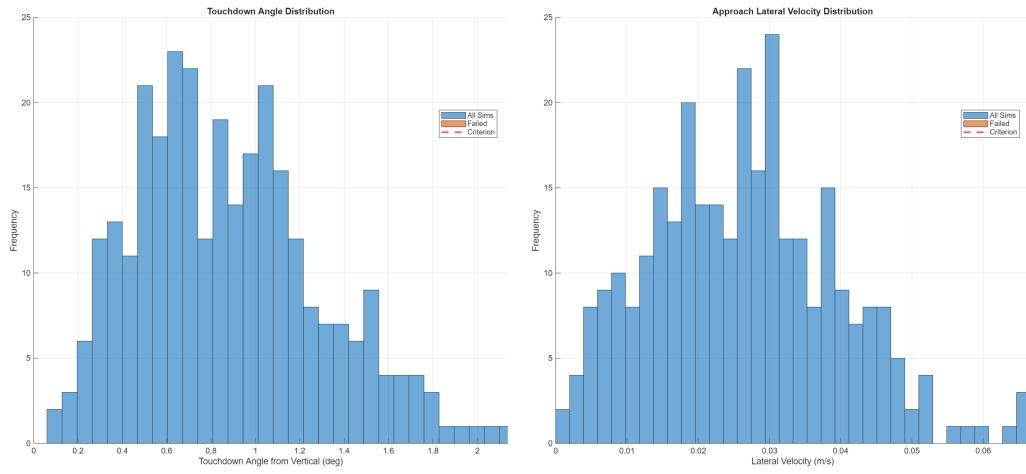


Figure 4.3: Approach Criteria Distributions

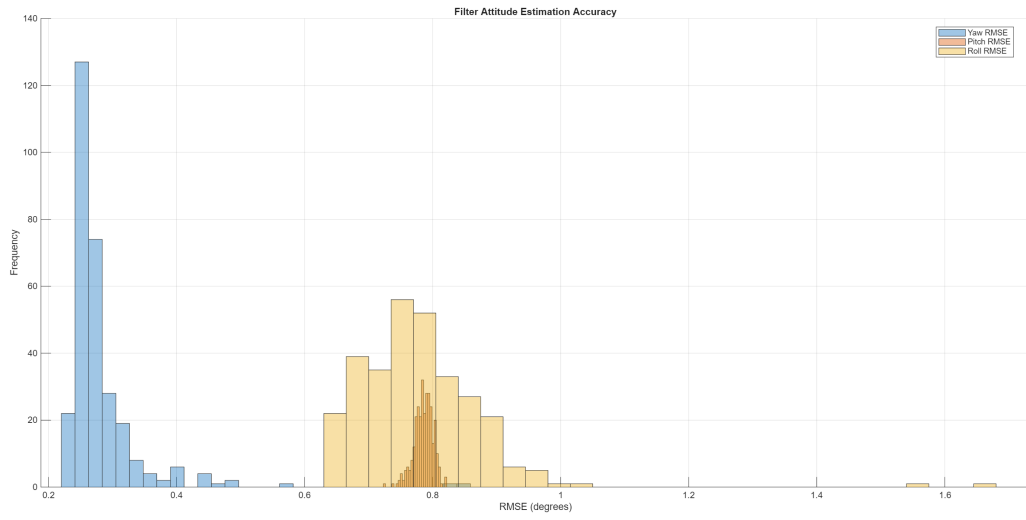


Figure 4.4: Filter Attitude RMSE

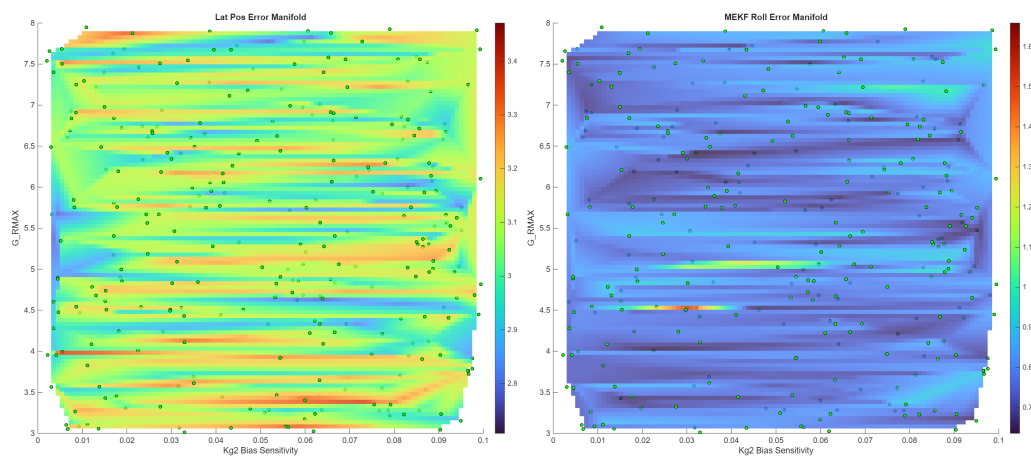


Figure 4.5: Stability Manifolds

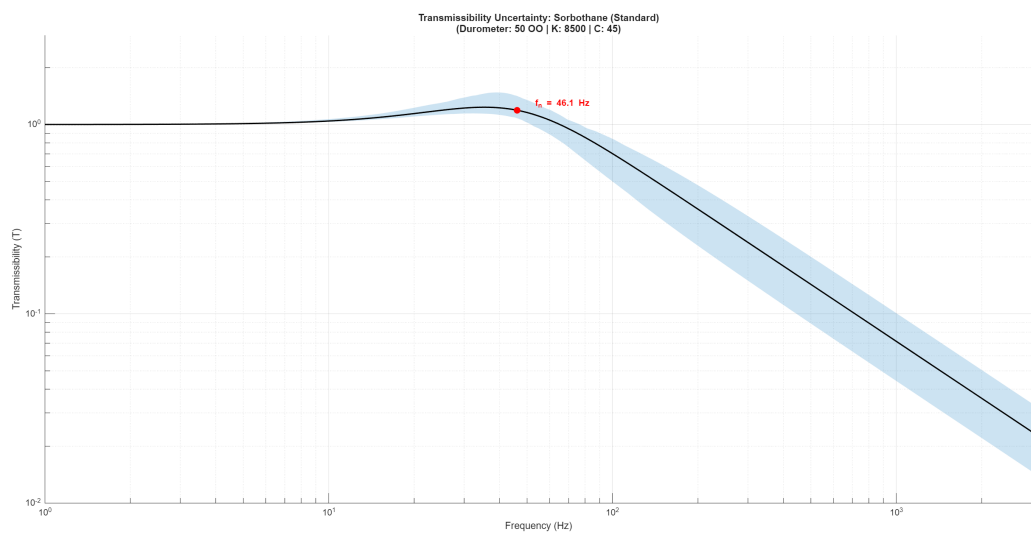


Figure 4.6: Grommet Transmissibility

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